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SUPERSEDING
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MILITARY SPECIFICATION

BROILER, CONVEYORIZED, INFRARED, ELECTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two styles of infrared conveyorized broilers.

1.2 Classification. Broilers shall be of the following styles and sizes, as specified (see 6.2):

Style I - Single conveyor.
Size 1 - 12-inch conveyor.
Size 2 - 18-inch conveyor.
Size 3 - 24-inch conveyor.

Style II - Two conveyors.

1.2.1 Part numbers. The specification number, style and size are combined to form the specification part numbers for the broilers covered by this specification. The part numbers for the broilers are established as follows:

Specification number	M MIL-B-43928
Style and size number (see 1.2.1.1)	XXX

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7310

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1.2.1.1 Style and size code. The style and size code are identified by a three digit number. The first digit identifies the style and the last two digits identifies the size as shown in table I.

TABLE I. Style and size to code number.

Styles and Sizes	Code Number
Style 1	
Size 1	101
Size 2	102
Size 3	103
Style II	204

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-B-11337 - Ranges, Packaging and Packing of.

FEDERAL

PP-B-2120 - Ground Beef Products, Frozen.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables of Inspection by Attributes.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for Control of Electromegnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

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(Copies of specifications, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A167 - Stainless and Heat Resisting Chromium - Nickel Steel Plate, Sheet and Strip.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 2 - Industrial Control Devices, Controllers and Assemblies.
ICS 6 - Enclosures for Industrial Controls and Systems.
MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

NATIONAL SANITATION FOUNDATION (NSF)

No. 4 - Commercial Cooking and Hot Food Service Equipment. Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, 3475 Plymouth Road, P. O. Box 1468, Ann Arbor, MI 48106.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 197 - Commercial Electric Cooking Appliances.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

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(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard commercial product. The broiler shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the broiler being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2).

3.3 Codes and standards. Broilers shall comply with the applicable requirements of NSF Standard No. 4 and UL No. 197.

3.4 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

3.4.1 Stainless steel sheet and strip. Stainless steel sheet and strip shall conform to any type in the 300 series of ASTM A167.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Design. Conveyor broilers shall be electrically operated and designed for counter use. The broiler body shall be open at the front and rear with conveyor(s) extending beyond the top hood at both ends. Radiant heating elements shall be located in the heat chamber above and below the conveyor(s). For the style I, size 1 broiler, the conveyor shall be 12 inches wide, the size 2 conveyor shall be 18 inches wide, and the size 3 conveyor shall be 24 inches wide. Style 1 broilers shall consist essentially of a body; one

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adjustable speed, motor driven, stainless steel conveyor; grease trough and receptacle; receiving pan; radiant heating elements; control panel. and legs. The style II broiler shall have two conveyors having a combined width of not less than 18 inches and not more than 24 inches. The narrower conveyor shall have a width of not less than 6 inches. Each conveyor for the style II broiler shall be individually motor driven and individually controlled. Other components of the style II broiler shall be as specified for style I broilers. Provisions shall be made on both styles to prevent seeping of grease into the drive mechanisms.

3.7 Performance requirements. The completed broiler shall operate without binding, erratic operation, excessive vibration, overheating, failure of any part or component, or any evidence of malfunction, parts separation, breakage, or any defect affecting the efficient and safe operation of the broiler, as demonstrated by the operational test specified in 4.5.1. The style I, size 1 broiler shall be capable of producing not less than 600 cooked hamburgers per hour. The style I, sizes 2 and 3, and the style II broilers shall produce not less than 700 cooked hamburgers per hour. The conveyor speeds shall be adjustable from 0.3 feet per minute to a speed which will produce the hamburger capacity specified. While cooking hamburgers or other meats on the primary conveyor, the style II broiler shall be capable of toasting buns at a variable and commensurate rate on the second conveyor. The hamburger patties shall be considered cooked when the internal temperature reaches a minimum of 160°F (see 4.5.2). This internal temperature shall be reached without charring the hamburger patty. Performance tests shall be as specified in 4.5.2.

3.7.1 Dimensions. When specified (see 6.2 and 6.6), broilers shall be within the dimensional limits shown in table II. When limited dimensions are required for broilers installed aboard ships (see 6.2), the width or length of the broiler, whichever is less, shall not exceed 30 inches. The height of broilers, including legs or base cabinet, shall be suitable for counter operation.

TABLE II Dimensions (inches maximum).

Measurement	Style I		Style II
	Size 1	Size 2	
Height 1/	25	25	25
Width 2/	20	30	30
Length	49	49	49

1/ Height from counter top including legs (overall).
2/ Handles excluded.

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3.7.1.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

3.8 Construction. All broilers components shall be fabricated of stainless steel specified in 3.4.1 including frame, body, conveyor, drive chains, grease trough, grease receptacle, heat reflector plate(s), access panels, legs, handles, and fasteners.

3.8.1 Body. The broiler body shall consist of a bottom plate, top cover, side panels, contoured end panels, heat reflector plate(s), top and bottom heating elements, and guides for grease drain trough and grease receptacle tray. Conveyor drive shaft mountings shall be suitably secured to rigid and reinforced components of the frame or body. The mountings shall be so constructed that the conveyor drive shafts can be removed easily with common tools such as a screwdriver. Means shall be provided in the body to maintain the conveyor(s) parallel to the heaters. The body shall be designed so that all parts requiring daily or periodic cleaning shall be removable without special skills or special tools. Means shall be provided to dissipate the heat from the heat chamber. The body shall be provided with bracing and reinforcing to form a rigid construction. A removable heat reflector plate shall be located above the top heating elements. All flanges located in the heat chamber shall be formed to insure grease runoff.

3.8.2 Conveyor(s). Broiler conveyors shall be as specified in 3.6, and shall be constructed of stainless steel wire links spaced not more than 3/4 inch apart. Conveyors shall be removable without the use of special tools. If required, provisions shall be made for adjusting and aligning the conveyor(s) and drive mechanisms. The conveyor(s) shall operate smoothly with no evidence of binding, erratic operation, unusual noise, excessive looseness or other evidence of malfunction, as demonstrated by the tests of 4.5.1.

3.8.3 Grease trough. The grease trough shall be of sufficient size to catch the grease drippings from the conveyor(s) and heat chamber sides. The grease trough shall be designed to preclude accidental ignition of drippings from the product being cooked on a continuous basis. The grease trough shall be removable and replaceable in the grease trough guides without any evidence of binding, as demonstrated by the tests of 4.5.1. Stops shall be provided on the guides so that the grease trough cannot accidentally slide out of position.

3.8.4 Grease receptacle. The grease receptacle shall be located under the grease trough and shall collect the grease from the trough. The grease receptacle shall be easily cleanable, and shall be removable and replaceable without evidence of binding as demonstrated by the test specified in 4.5.1.

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3.8.5 Receiving pan. The receiving pan shall be located at the discharge end of the conveyor(s) to receive the cooked product. The receiving pan shall be closed on three sides and open on the side adjacent to the heat chamber to allow grease to drain onto the grease trough. The receiving pan shall be easily cleanable, easily removable and replaceable without binding when tested as specified in 4.5.1.

3.8.6 Supports. Broilers shall be supported on a suitable cabinet or on four adjustable stainless steel legs located at the bottom four corners of the broiler. Each leg adjustment shall be not less than one inch. The legs shall be tubular or angular in shape with a nominal thickness of not less than 0.0625 inch. For shipboard use, (see 6.2), each support bottom shall be tapped to receive a 3/8-16 hold-down bolt or may be provided with brackets, angles or other hold-down means for securing.

3.9 Electrical system. Unless otherwise specified (see 6.2), broilers shall be equipped for operation on a 440 volts (V), 60 Hertz (Hz), 3-phase system except that power supply for controls and motor shall be 208-230/120V, 3-phase. Broilers shall have provisions for permanent electrical connection.

3.9.1 Wiring. Electrical wiring design and protection practices shall be in accordance with NFPA 70 and UL 197, as applicable.

3.9.2 Control panel. The control panel may be located at the load end or at either side of the broiler. The control panel shall be designed to conform to the requirements of NEMA ICS-2 controller and ICS-6 type 2 enclosure. The control panel shall be protected from grease drippings. The control panel shall be provided with conveyor speed control(s), heating element switches, indicating light(s) and fuse unit(s). When specified for shipboard use (see 6.2), the control panel shall be located at the right side of the broiler when viewing the broiler from the conveyor discharge end.

3.9.3 Conveyor speed control. Styles I and II broilers shall be provided with one and two conveyor speed controls, respectively. Control(s) shall be of the potentiometer type. The control knob(s) shall be provided with numerical dial settings from 10 to 100 (see 4.5.1). Conveyor speed shall increase as dial setting is increased.

3.9.4 Switches. Broilers shall be provided with switches to control the electric circuit to the upper and lower heating unit assemblies.

3.9.5 Indicating light. Styles I and II broilers shall be provided with one and two indicator lights, respectively. Indicator light(s) shall glow whenever the conveyor drive motor electrical system is energized (see 4.5.1).

3.9.6 Heating unit assembly. One heating unit assembly shall be located above the product and one shall be located below the product. Each heating unit assembly shall be provided with electrical heated intra-red radiant heaters. Heating unit assemblies or heating elements, as applicable, shall be removable and replaceable without binding when tested as specified in 4.5.1. Heater unit electrical leads shall be protected, routed, and secured to prevent electrical leads from rubbing against moving components.

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3.9.7 Motors. Motor overload protection shall be provided in accordance with NFPA 70 requirements. Conveyor motor wiring shall be adequately protected from overheating in excess of their insulation rating as defined in NFPA 70. Motors shall be totally enclosed, continuous duty type and conform to the requirements of NEMA MG1.

3.9.8 Electromagnetic compatibility. When specified (see 6.2), the units shall be designed and equipped for electromagnetic compatibility in accordance with class C3, UM05, Group I of MIL-STD-461 (see 4.5.4).

3.10 Data name plates. Data name plates shall be made of minimum 20 gauge corrosion-resisting metal and attached to each broiler by rivets, screws or welding in such a manner as to meet the applicable National Sanitation Foundation sanitary requirements for this equipment. The plate shall contain the following information stamped, engraved or applied by photosensitive means:

National Stock Number
Procurement Instrument Identification Number
Manufacturer's Name, Address, Phone Number
Manufacturer's Model Number

Each plate shall be placed so that it is readily visible to the operator during normal operating use. Each plate shall be placed in a manner so as to not adversely affect the life and utility of the item.

3.11 Instruction plates. The broiler shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter. The plate shall be of sufficient overall size to include the following information in legible characters:

WARNING

High voltages are present in the unit.
Make sure all input power is OFF, prior to using cleaning agents or solutions in the broiler or for parts replacement and servicing.

CAUTION

DO NOT clean or wash any heating elements.
Heating elements are self-cleaning and are completely sterile within 60 seconds of operation.

3.12 Operation. Each broiler heating element, conveyor and mechanism, conveyor speed control, switches and indicator light(s) shall operate satisfactorily when tested as specified in 4.5.1.

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3.12.1 Incline operation for shipboard use (see 6.2). Each broiler shall operate satisfactorily when inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other, with no spillage of fluid or product when tested as specified in 4.5.3.

3.13 Repair and maintenance. Provision shall be made to allow access to permit adjustment, repair, or replacement of electrical and other components.

3.14 Accessory parts. Each broiler shall be provided with the following accessories:

Style I
- Size 1 - One set of spare heating elements
- One 12 inch conveyor
- Size 2 - One set of spare heating elements
- One 18 inch conveyor
- Size 3 - One set of spare heating elements
- One 24 inch conveyor

Style II - One set of spare heating elements
- One of each size conveyor

3.15 Finish. All surfaces shall conform to NSF Standard No. 4. Broilers shall be smooth on all surfaces and edges and shall be free from discoloration or stains.

3.16. Workmanship.

3.16.1 Metal fabrication. The metal used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the metal to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.16.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.16.3 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

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3.16.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be one of the following:

a. A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished broiler, or
b. A certification for the broilers issued by NSF under their special one-time contract evaluation/certification service, or
c. A test report, acceptable to the contracting officer with the advice of the Army Surgeon General, from an independent testing laboratory indicating that the items have been tested and conform to the specified NSF Standard No. 4.

4.1.1.2 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory indicating that the broilers have been tested and conform to UL 197.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one broiler when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4 and test of 4.5.1 and 4.5.2 and, when applicable, 4.5.3 and 4.5.4; and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one broiler. All broilers of the same classification offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot shall be screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for visual examination. Examinations shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent for major defects and 6.5 percent for total defects, expressed in terms of defects per hundred units (DHU).

4.3.2 Sampling for dimensional examination. Examination shall be based on inspection level S-2 and AQL of 4.0 DHU's.

4.3.3 Sampling for tests. Each broiler shall receive the operational test of 4.5.1. For the tests of 4.5.2 , 4.5.3, and 4.5.4, one production unit shall be randomly selected for the applicable tests.

4.3.4 Sampling for preparation for delivery inspection. Sampling for preparation for delivery inspection shall be in accordance with the requirements of section 4 of MIL-R-11337.

4.4 Examinations.

4.4.1 Visual examination. Each broiler shall be examined for defects listed in table III. Each attribute within each classification of multiple defects shall constitute one defect.

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TABLE III Classification of defects.

Classification	Defects	Requirement paragraph
Critical:		
1	Any condition which is hazardous or unsafe for individuals using the broiler.	3.3
Major:		
101	Broiler not of correct classification specified	1.2
102	Broiler not in accordance with manufacturer's standard product, except for changes necessary to comply with specification requirements.	3.1
103	Broilers not in conformance to industry standards.	3.3
104	Component missing.	3.4
105	Materials not as specified.	3.4, 3.4.1
106	Interchangeability not provided for.	3.5
107	Design not as specified.	3.6
108	Body not as specified.	3.8.1
109	Conveyor not aligned.	3.8.2
110	Grease trough not as specified.	3.8.3
111	Grease receptacle not as specified.	3.8.4
112	Receiving pan not as specified.	3.8.5
113	Supports not as specified.	3.8.6
114	Electrical system not as specified.	3.9
115	Control panel not as specified.	3.9.2
116	Conveyor speed control missing.	3.9.3
117	Electrical switches missing.	3.9.4
118	Indicator lights missing or do not function.	3.9.5
119	Heating unit assembly not as specified.	3.9.6
120	Motor not as specified.	3.9.7
121	Electromagnetic compatibility not provided for when required.	3.9.8
122	Data name plates missing, incomplete, not legible, not as specified.	3.10
123	Instructions missing, incomplete, not legible.	3.11
124	Instruction plate not as specified.	3.11
125	Access not provided for.	3.13
126	Accessory parts missing or not as specified.	3.14, 3.8.2, 3.9.5
127	Surfaces not smooth and not in compliance with NSF Standard No. 4.	3.15
Minor:		
201	Parts to body not removable for cleaning.	3.8.1
202	Conveyor(s) do not operate smoothly.	3.8.2
203	Supports loose.	3.8.6
204	Permanent electrical connection not provided for	3 9
205	Conveyor speed control(s) not as specified.	3.9.2, 3.9.3
206	Electrical switches not as specified.	3.9.4
207	Heating unit assembly or element binds when removed.	3.9.6
208	Finish not free of discoloration or stain.	3.15
209	Metal fabrication, welding, and fastening devices not as specified.	3.16.1, 3.16.2, 3.16.3

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4.4.2 Dimensional examination. The broilers shall be examined for compliance with dimensions specified. Any dimension not within the specification tolerance shall be a defect.

4.5 Tests.

4.5.1 Operational test. Connect the broiler to an electrical power source having the same characteristics specified in the contract and check to assure that all moving components function as intended. Observe that indicator light(s) glow when the conveyor motor electrical systems are activated and goout when power is turned "OFF". While conveyor(s) are operating, start and stop conveyor motor(s) at least ten (10) times. Observe indicator light(s) for compliance with 3.9.5. Check to assure that conveyor speed control(s) operate through the complete range to determine compliance with 3.9.3. Change the speed control settings at least ten (10) times and observe conveyor speed for compliance with 3.9.3. Observe all components for compliance with 3.7. During conveyor operation, check for conformance with 3.8.2. Energize top heating element switch and observe that the heating elements glow. Turn top switch "OFF" and observe that top heating elements go out. Repeat this procedure for the lower heating elements to determine conformance with 3.12. Remove and replace the upper and lower heating units, grease trough, grease receptacle and receiving pan to determine conformance with 3.8.3, 3.8.4, and 3.8.5 respectively. Any nonconformance with specified requirements shall constitute failure of this test.

4.5.2 Performance test. Using the same electrical power source in 4.5.1, preheat the broiler in accordance with the manufacturer's standard procedure. During preheat period, turn heating switches and conveyor motor speed control(s) "ON". After broiler has reached proper preheat temperature, load the conveyor(s), as applicable, with hamburger patties four inches in diameter and not less than 3/8-inch in thickness, conforming to type III of PP-B-2120. Fat content of the hamburger shall not exceed 28 percent. Hamburger patties prior to test shall be 35 to 40°F. Grill hamburger patties until internal temperature conforms to 3.7. The internal temperature shall be taken at the center of the hamburger patty. The temperature indicating device shall be of the quick reading type or a thermocouple probe. After specified hamburger patty internal temperature has been attained, repeat the temperature reading procedure on additional patties for thirty (30) minutes. During this test, hourly rate shall be checked to determine conformance with 3.7. Any nonconformance shall constitute failure of this test.

4.5.3 Inclined operational test. This test is applicable for broilers intended for shipboard use. For this test the grease receptacle and receiving pan shall be filled to 75 percent capacity with water. After the test in 4.5.2 is completed, position the broiler with the base set at an angle of 15 degrees, then operate broiler for 5 minutes each at each side of the vertical in each of two vertical planes at right angles to each other. At each of these positions, continuously load conveyor with hamburger patties for 5 minutes to determine conformance with specified requirements of 3.12.1. Any nonconformance with specified requirements shall constitute failure of this test.

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4.5.4 Electromagnetic compatibility test. When electromagnetic compatibility is required (see 3.9.8), the initial production unit shall be Tested by the supplier in accordance with requirements of UMO5, MIL-STD-461 and applicable test method UMO5 of MIL- STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Compatibility Test Report and the requirements meet MIL-STD-461. Inability of the broiler to pass the electromagnetic compatibility test shall constitute failure of the unit.

4.6 Preparation for delivery inspection. The inspection of the preservation, packaging, packing, and marking shall be in accordance with the requirements of section 4 of MIL-R-11337.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, packing, and marking shall be in accordance with the requirements of MIL-R-11337 with the level of preservation and packaging and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The broilers covered by this specification are intended for use in military kitchens, galleys, and fast food preparation facilities for broiling of various food products.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Style and size broiler required (see 1.2)
c. When first article is required (see 3.2).
d. When required for shipboard use (see 3.8.6 and 3.12.1).
e. When other than 440V, 60 Hz, 3-phase electric power is required (see 3.9).
f. When electric power characteristics for control(s) and motor shall be other than as specified (see 3.9).
g. When control panel location is required for shipboard (see 3.9.2).
h. When electromagnetic compatibility is required (see 3.9.8).
i. Level of preservation and packaging and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

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6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Subject term (key word) listing.
Appliance, commercial.
Broiler, conveyorized, electric.
Food service equipment.

6.6 Information to contracting officers. The limiting dimensions shown in table II are included to enable procurement of replacement broilers in existing messing facilities. Additional sources of supply, with resultant decrease in cost, may occur if limiting dimensions are not specified, as applicable to new or remodeled food service lines.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Navy - YD
Air Force - 99

Review Activities:
Navy - SA
Air Force - 99
DLA - GS

User Activity:
Navy - MC

Preparing Activity:
Navy - YD

(Project 7310-0699)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-43928A	BROILER, CONVEYORIZED, INFRARED, ELECTRIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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